

Are Viral Exposures Inducing Accelerated Ageing?

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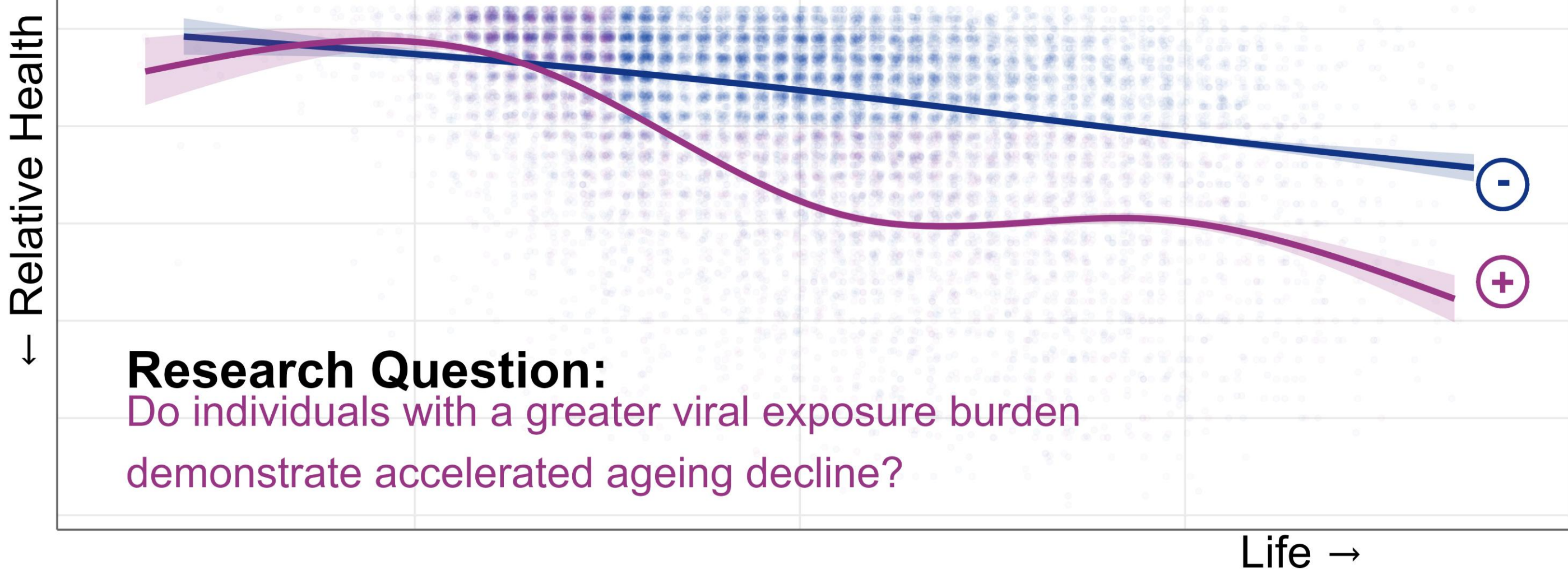
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1. Research Background

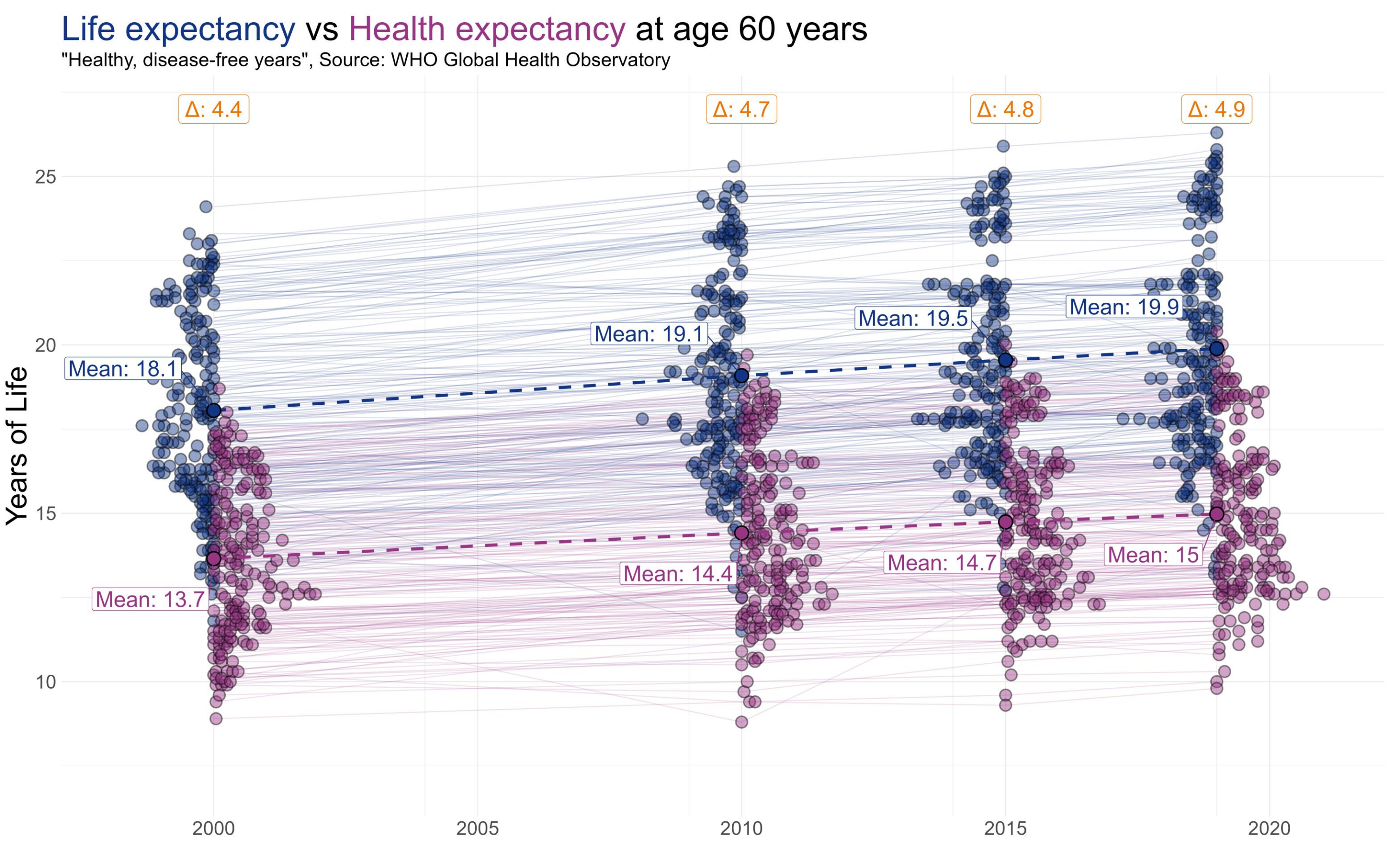
Global Ageing Population: Individuals aged 60+ is projected to double by 2050, from 1 billion to 2.1 billion.
That's good news.

The Healthspan-Lifespan Problem: Divergence between biological and chronological ageing is widening, meaning larger proportions of later life in poor health and straining healthcare systems.
That's bad news.



Persistent Viral Drivers: Persistent viral infections drive chronic inflammation and T-cell exhaustion, but their long-term impact on differential ageing trajectories remains unclear.

Core Hypothesis: Viral exposures accelerate biological and immune ageing in an age-dependent manner, contributing towards later life morbidity burden.

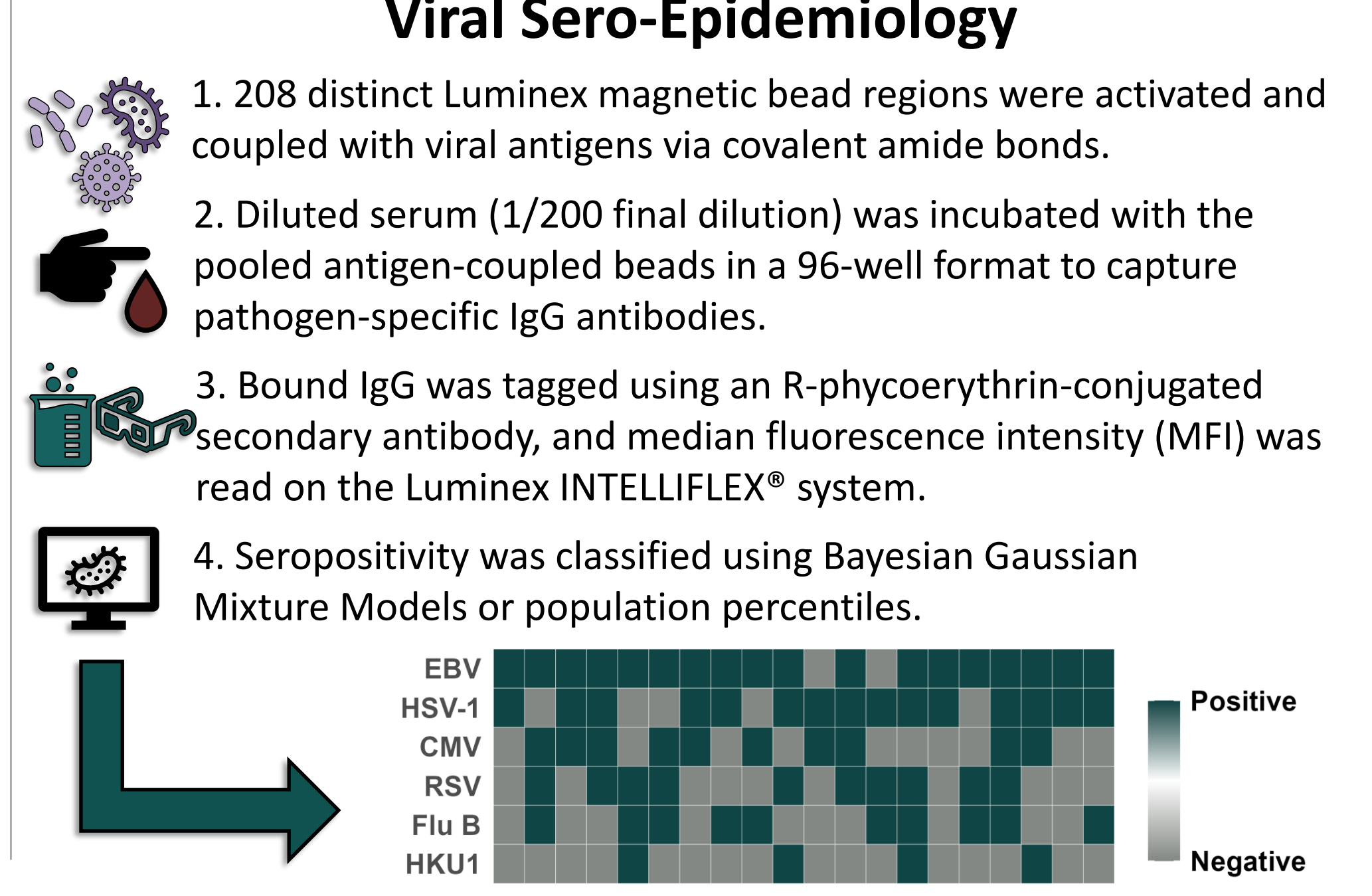
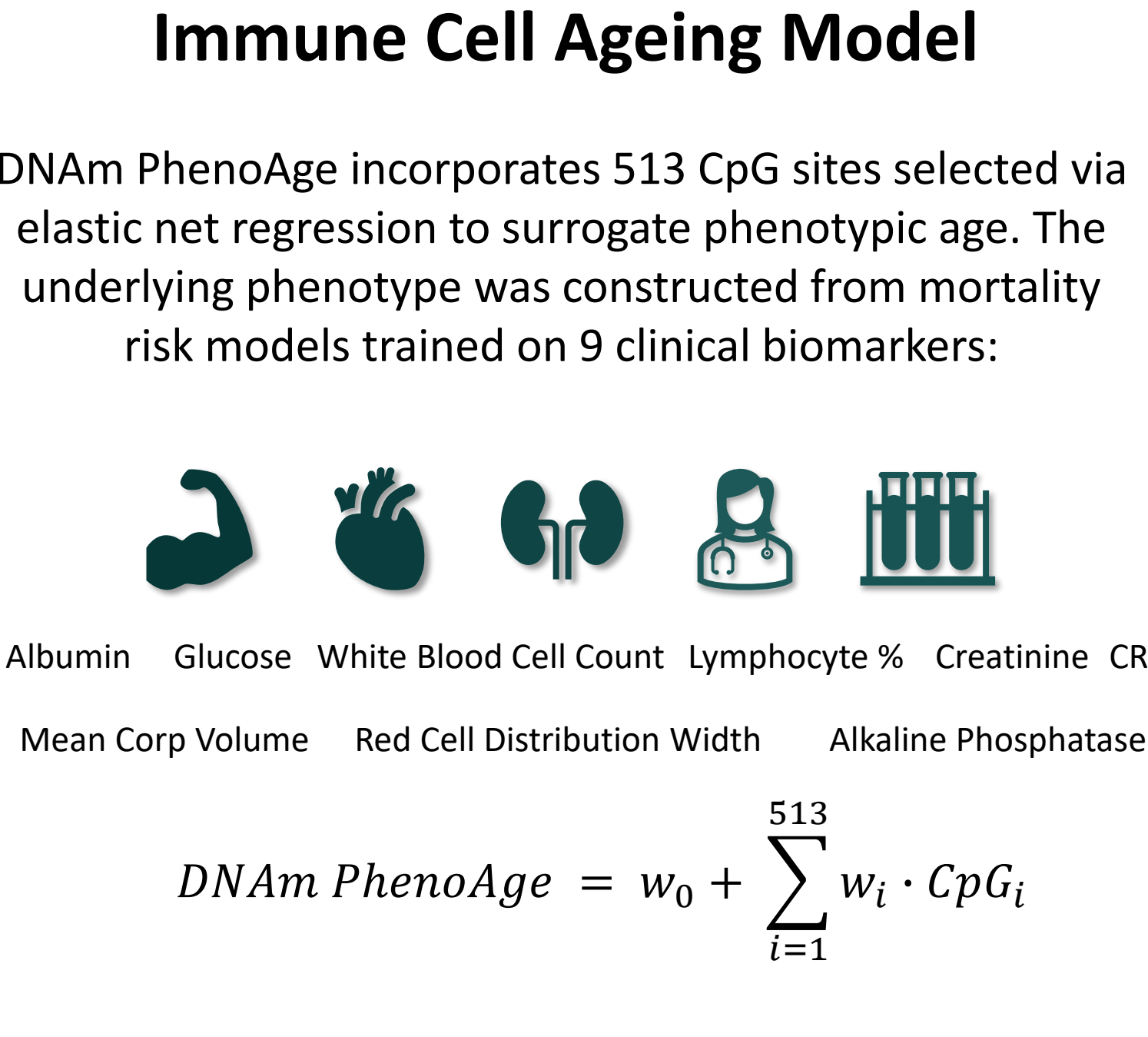
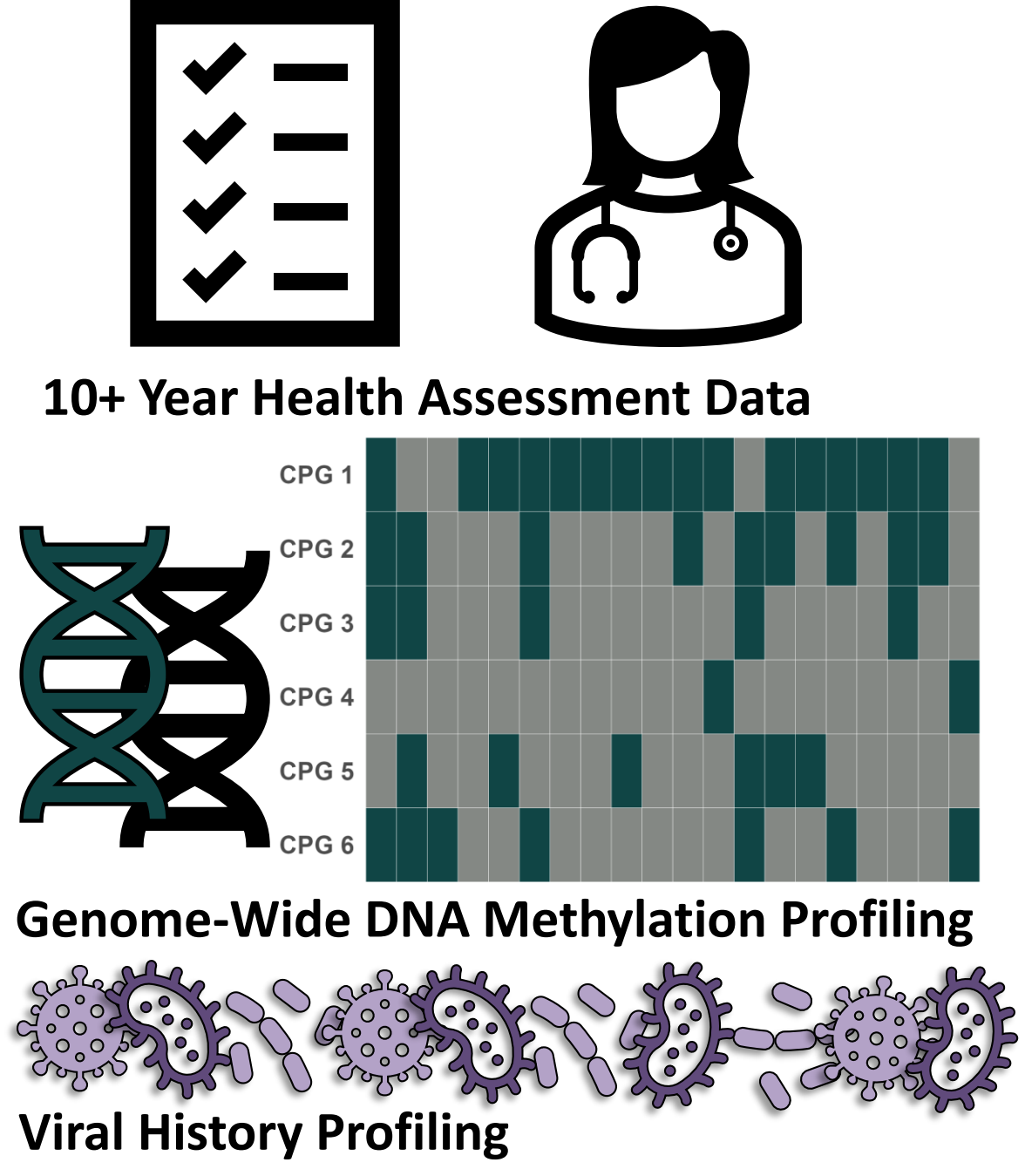


2. Methods

The Milieu Intérieur Cohort
The *Milieu Intérieur* cohort is a deeply phenotyped, population-based study of 1,000 healthy individuals aged 20 to 69 (stratified by age and sex) living in France.

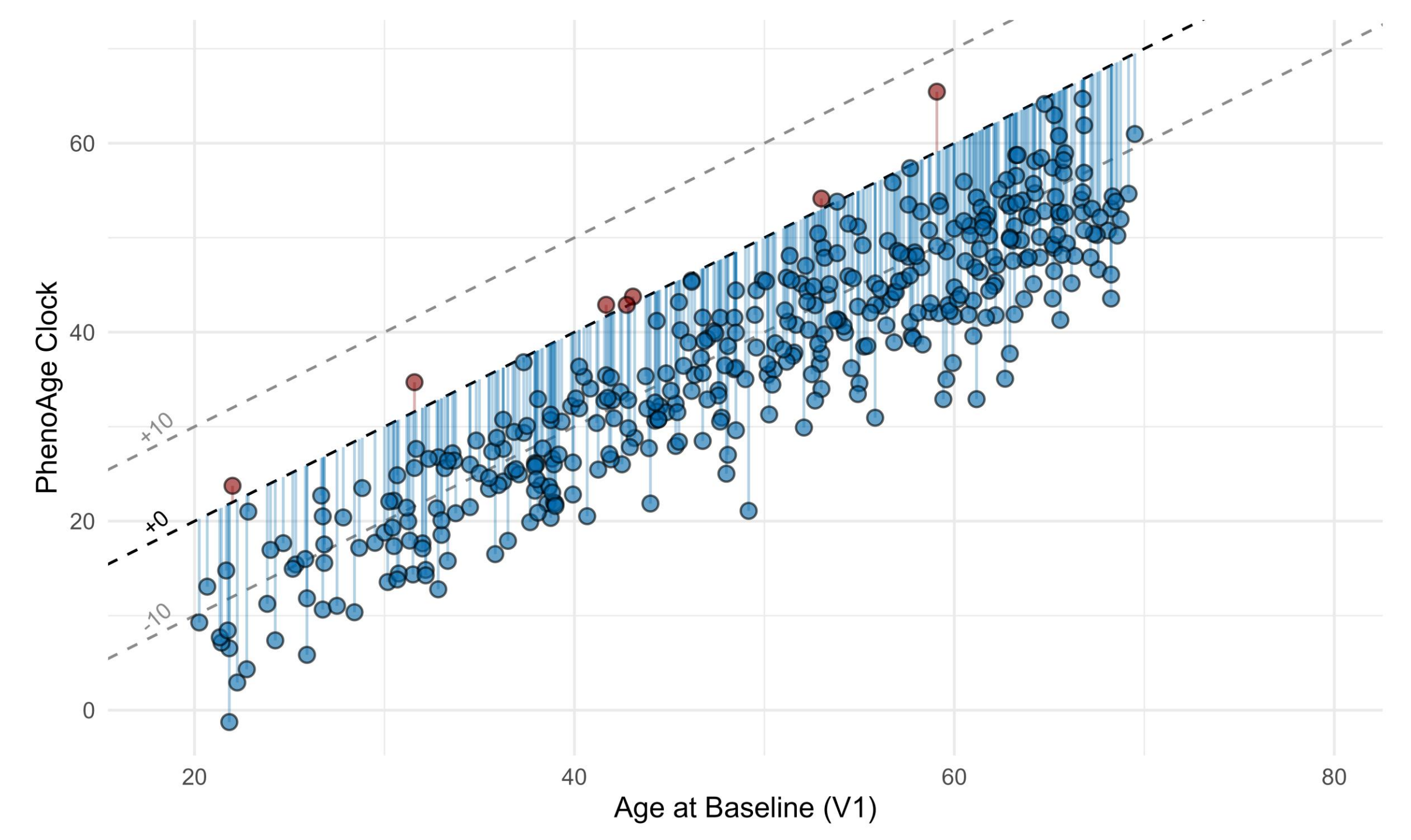
Milieu Intérieur's goal is to establish the boundaries of a healthy immune response and identify the genetic, epigenetic, and environmental factors – including chronic viral exposure – that drive inter-individual variability and immune decline across the lifespan.

N = 1,000
500 x Healthy Men & Women

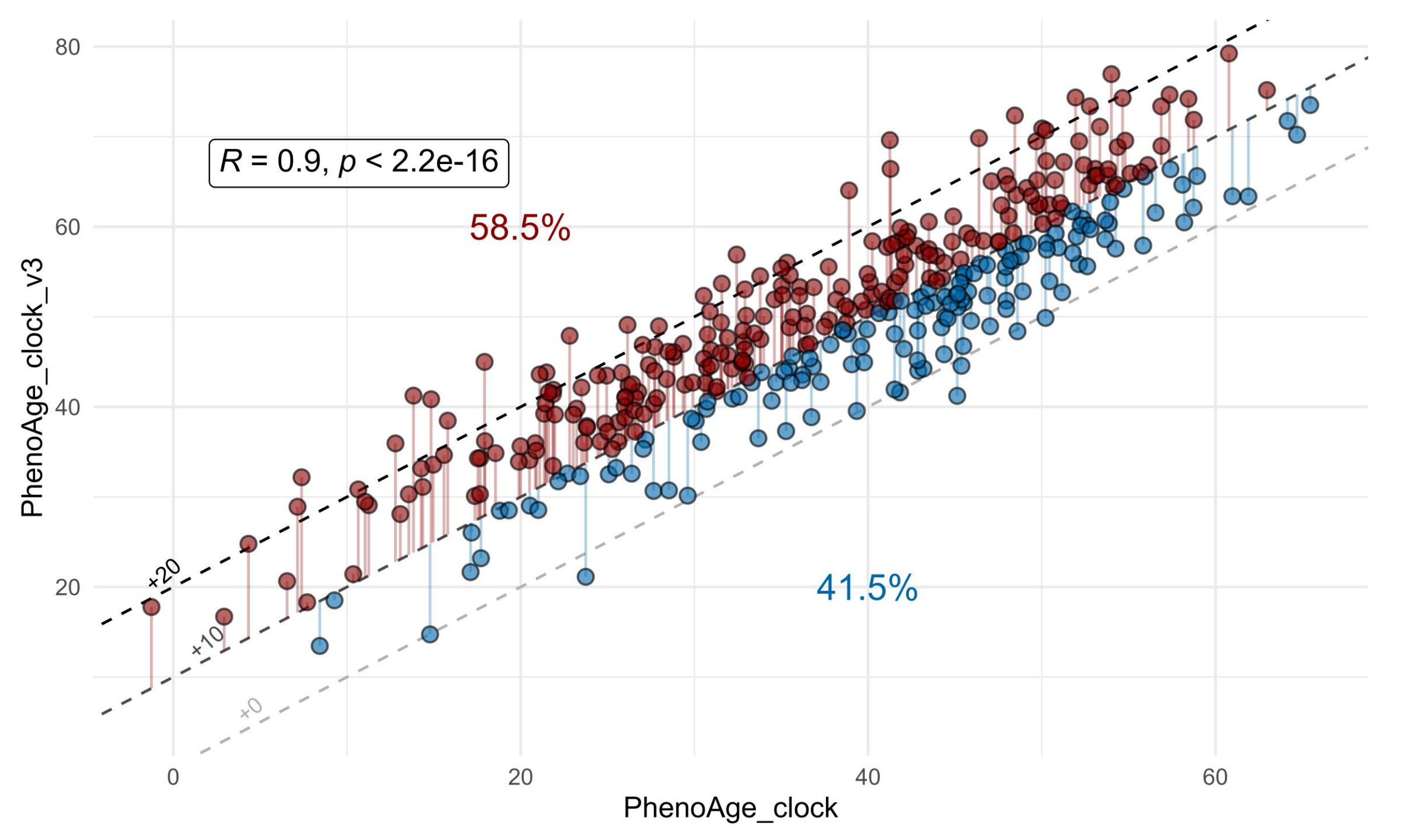


3. Results

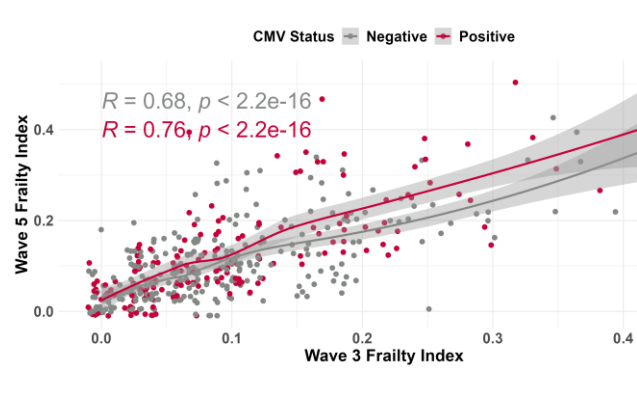
Milieu Intérieur Cohort Immune Methylomes Appear Around 10 Years Biologically Younger Than Expected.
● Ageing Slower ● Ageing Faster



After 10 years, some individuals have aged more than others, with the population average being around +10.
● Ageing Slower ● Ageing Faster



github.com/MattMcElheron/McElheron_ISI_Poster
Follow the QR code or link to see figure animations, further visualizations, explanations of caveats, and subsequent findings using a nationally representative cohort.



10-year age acceleration is strongly associated with compositional changes, not viral exposures, in healthy individuals.
● Ageing Slower ● Ageing Faster

